

W2 - 1.2 Functions and Function Notation

MCR3U

1) For each function, determine $f(4)$, $f(-5)$, and $f\left(-\frac{2}{3}\right)$.

a) $f(x) = \frac{2}{5}x + 11$

b) $f(x) = 3x^2 + 2x + 1$

c) $f(x) = 2(x + 4)^2$

d) $f(x) = -6$

e) $f(x) = \frac{1}{x}$

f) $f(x) = \sqrt{x + 5}$

2) If $f(x) = x^2 + 2$, state the following.

a) $f(1)$

b) $f(0)$

c) $f(2)$

d) $f(-2)$

e) $f(3)$

f) $f\left(\frac{1}{2}\right)$

3) State $f(4)$ for each of the following functions.

a) $f(x) = 4 + 5x$

b) $f(x) = x^2 - 6$

c) $f(t) = 9 - t$

d) $f(x) = 10$

e) $f(z) = z^3$

f) $f(x) = 8(5 - x)$

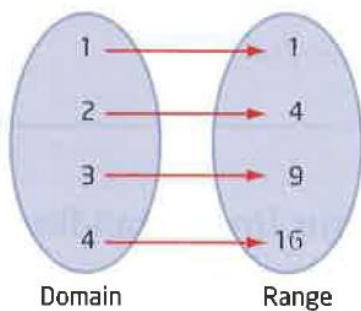
g) $f(x) = \frac{1}{x}$

h) $f(x) = \sqrt{13 - x}$

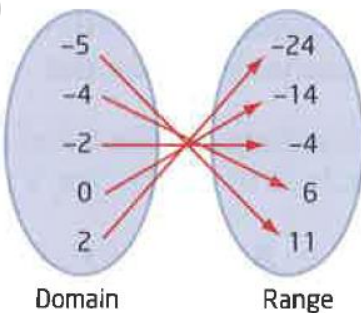
i) $f(t) = \frac{1}{t^2}$

4) Write the ordered pairs associated with each mapping diagram. Then state if the relation is a function.

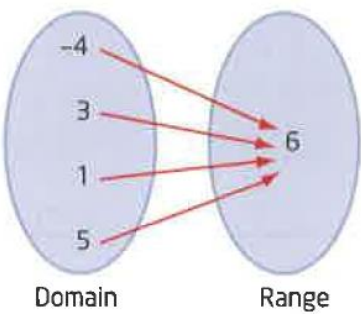
a)



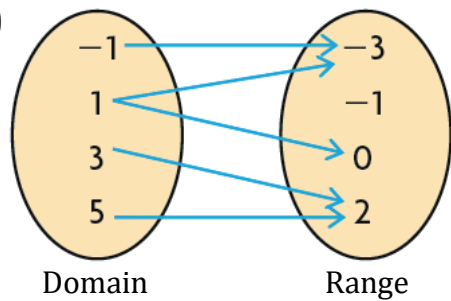
b)



c)



d)



5) Show each set of data in a mapping diagram. Then state if the relation is a function.

a) $\{(1, 4), (2, 1), (3, -2), (4, -5), (5, -8), (6, -11), (7, -14), (8, -17)\}$

b) $\{(-3, 4), (-2, -1), (-1, -4), (0, -5), (1, -4), (2, -1)\}$

c) $\{(-5, 6), (-4, 9), (-3, 1), (-5, -6), (1, -2), (3, 8), (8, 8)\}$

d) $\{(9, 9), (7, 9), (5, 9), (3, 9)\}$

6) State the domains of the following functions

a) $f(x) = \sqrt{8 - x}$

b) $f(x) = \frac{x^2+3}{(x-1)(x+3)}$